Pentium 4 ITLB, Branch Prediction & Hyper-Threading MPCA Module 6

> **Definition:** Pentium 4 integrates an **Instruction Translation Lookaside Buffer (ITLB)**, advanced **Dynamic Branch Prediction**, and **Hyper-Threading (HT) Technology** (Simultaneous Multithreading) to maximize execution unit utilization.

---

1. Detailed Technical Explanation

1. Instruction Translation Lookaside Buffer (ITLB)

- A high-speed associative hardware cache that caches virtual-to-physical page address translations for instruction fetch.

- Elir

---

2. Advanced Dynamic Branch Prediction

- Because the NetBurst pipeline is **20 stages deep**, a branch misprediction penalty results in flushing 20 stages of instructions (a massive performance penalty).

- Net

---

3. Hyper-Threading Technology (HT Technology)

Hyper-Threading is Intel's hardware implementation of **Simultaneous Multithreading (SMT)** that allows a single physical processor core to appear as **two logical processors** to the operating system.

```

WITHO

WITH HYPER-THREADING (Simultaneous Multithreading):

Logica

### Architecture of Hyper-Threading:

1. **Duplicated Architectural State:**

- Re
**dupli

---

## 4. Must-Write Points for Exams

- Hyper-Threading presents one physical core as two logical cores to the OS scheduler.
```

- Arc

---

5. Quick Recall Flow

```

ITLB F
(Duplic
```